

- 7 (a) *either* forms of same element
 or atoms / nuclei with same number of protons M
 atoms / nuclei contain different numbers of neutrons A1
 (use of 'element' rather than atoms / nuclei scores max 1 mark)
- (b) (i) decay is not affected by environmental factors B1 [1]
 (allow two named factors)
- (ii) *either* time of decay (of a nucleus) cannot be predicted
 or nucleus has constant probability in a given time B1 [1]
- (c) $^{185}_{75}\text{Re}$ B1
 either $^0_{-1}\text{e}$ *or* $^0_{-1}\beta$ B1 [2]

[Total: 6]